

Exercises-5

1. $x = (2, -1, 0, -3)$, $y = (1, -2, -1, 5)$, $z = (1, 2, -5, 2)$

$$\begin{aligned} \text{(i)} \quad 2x - 5y &= 2(2, -1, 0, -3) - 5(1, -2, -1, 5) \\ &= (4, -2, 0, -6) - (5, -10, -5, 25) \\ &= (-1, 8, 5, -31) \end{aligned}$$

ચાલિશૂના બચે ,

$$\begin{aligned} \text{(iv)} \quad x \cdot y &= (2, -1, 0, -3) \cdot (1, -2, -1, 5) \\ &= 2 + 2 + 0 + (-15) \\ &= 4 - 15 = -11 \end{aligned}$$

ચાલિશૂના બચે ,

$$\text{(v)} \quad \|x\| = \sqrt{(2)^2 + (-1)^2 + 0^2 + (-3)^2} = \sqrt{4+1+0+9} = \sqrt{14}$$

$$\|y\| = \sqrt{1^2 + (-2)^2 + (-1)^2 + 5^2} = \sqrt{1+4+1+25} = \sqrt{31}$$

$$\|z\| = \sqrt{1^2 + 2^2 + (-5)^2 + 2^2} = \sqrt{1+4+25+4} = \sqrt{34}$$

2. Given, $x = (1, \lambda, -2, 5)$ and $\|x\| = \sqrt{39}$

$$\|x\| = \sqrt{39}$$

$$\Rightarrow \lambda^2 = 9$$

$$\Rightarrow \sqrt{1^2 + \lambda^2 + 4 + 25} = \sqrt{39}$$

$$\Rightarrow 1^2 + \lambda^2 + 29 = 39$$

$$\Rightarrow \lambda^2 = 39 - 30$$

3. (i) Since x, y orthogonal,

$$(x, y) = 0 \quad (x, y) = 0 \quad (x, y) = 0$$

$$(x, y) = 0 \quad (x, y) = 0 \quad (x, y) = 0$$

$$\Rightarrow \lambda + \lambda^2 + 4 - 6 = 0$$

$$\Rightarrow \lambda^2 + \lambda - 2 = 0$$

$$\Rightarrow \lambda^2 + 2\lambda - \lambda - 2 = 0$$

$$\Rightarrow \lambda(\lambda + 2) - 1(\lambda + 2) = 0$$

$$\Rightarrow (\lambda - 1)(\lambda + 2) = 0$$

$$\therefore \lambda = -2, 1$$

वाक्यगुणा वक्तव्य 1

$$||x|| = \sqrt{1^2 + 0^2 + 2^2 + (-1)^2} = \sqrt{6} = ||x||$$

$$||y|| = \sqrt{2^2 + 1^2 + 3^2 + \beta^2} = \sqrt{14 + \beta^2} = ||y||$$

$$\therefore xy = 0$$

$$\Rightarrow (1, 0, 2, -1) \cdot (2, 1, 3, \beta) = 0 \Rightarrow 2 + 0 + 6 - \beta = 0$$

$$\Rightarrow 2 + 0 + 6 - \beta = 0$$

$$\Rightarrow \alpha - \beta = -8 \quad \text{--- (I)}$$

and,

$$xz = 0$$

$$\Rightarrow (1, \alpha, 2, -1) \cdot (\beta, 1, -4, 2) = 0 \Rightarrow \beta + \alpha - 8 - 2 = 0$$

$$\Rightarrow \beta + \alpha - 8 - 2 = 0$$

$$\Rightarrow \alpha + \beta = 10 \quad \text{--- (II)}$$

from ⑪ - ⑩

$$2\beta = 18 \quad \therefore \beta = 9.$$

from ⑩, $\alpha = -8 + 9 = 1$

$$\therefore \alpha = 1, \beta = 9.$$

⑤. $x = (6, 3, -2, -4, 1)$, $y = (3, -1, 0, -7, -2)$

$$\begin{aligned} \therefore d(x, y) &= \sqrt{(6-3)^2 + (3-(-1))^2 + (-2-0)^2 + (-4+7)^2 + (1+2)^2} \\ &= \sqrt{3^2 + 4^2 + 4 + 3^2 + 3^2} \\ &= \sqrt{9+16+4+9+9} = \sqrt{47} \end{aligned}$$

⑥ Given, $\|x-y\| = 6$ where $x = (4, \lambda, 1, -3)$,
 $y = (3, -1, 6, -2)$

$$\therefore x-y = (4, \lambda, 1, -3) - (3, -1, 6, -2) = (1, \lambda+1, -5, -1)$$

$$\therefore \|x-y\| = 6$$

$$\Rightarrow \sqrt{1^2 + (\lambda+1)^2 + (-5)^2 + (-1)^2} = 6$$

$$\Rightarrow 1 + (\lambda+1)^2 + 25 + 1 = 36$$

$$\Rightarrow (\lambda+1)^2 = 9$$

$$\Rightarrow \lambda^2 + 1 = \pm 3$$

$$\Rightarrow \lambda = \pm 3 - 1$$

$$\therefore \lambda = (3-1), (-3-1)$$

$$= 2, -4$$

7. Given $P(3, \lambda, -2)$ and $Q(5, 3, 4)$

$$\therefore \vec{PQ} = (5-3, 3-\lambda, 4+2) = (2, 3-\lambda, 6)$$

Since $\vec{PQ}$ is orthogonal to $(4, -3, 2)$

$$\therefore (2, 3-\lambda, 6) \cdot (4, -3, 2) = 0$$

$$\Rightarrow 8 - 9 + 3\lambda + 12 = 0$$

$$\Rightarrow 3\lambda = -11$$

$$\therefore \lambda = \frac{-11}{3}$$

$$(1-\epsilon-\epsilon) \cdot (1-\epsilon) = 1 \therefore$$

$$1-\epsilon-\epsilon =$$

$$d = \|B - x\|$$

$$d = \sqrt{(1-\epsilon)^2 + (\epsilon-\epsilon)^2 + (1+\epsilon)^2 + \epsilon^2}$$

$$2\epsilon = 1 + \epsilon^2 + (1+\epsilon)^2 + 1$$

$$\epsilon = \frac{1}{2}(1+\epsilon)$$

$$\epsilon \pm = 1 + \epsilon$$

$$1 - \epsilon \pm = 1$$

$$8. \quad u = (5-3i, 2+3i) \text{ and } v = (3+5i, 1-2i) \Rightarrow u \cdot v =$$

$$\begin{aligned} \text{(i)} \quad 2u + 3v &= 2(5-3i, 2+3i) + 3(3+5i, 1-2i) \\ &= (10-6i, 4+6i) + (9+15i, 3-6i) \\ &= (19+9i, 7+0i) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 2iu &= 2i(5-3i, 2+3i) = (10i-6i^2, 4i+6i^2) \\ &= (10i+6, 4i-6) \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad u \cdot v &= (5-3i, 2+3i) \cdot (3+5i, 1-2i) \\ &= (5-3i)(3+5i) + (2+3i)(1-2i) \\ &= \cancel{15+25i-9i-15i^2} + \cancel{2-4i+3i-6i^2} \\ &= \cancel{47+15i-24i} = \cancel{47-9i} \\ &= (5-3i)(3-5i) + (2+3i)(1+2i) \\ &= 15-25i-9i+15i^2 + 2+4i+3i+6i^2 \\ &= 17-27i+21i^2 = 17-27i-21 = -4-27i \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad \|u\| &= \sqrt{(5-3i)(5+3i) + (2+3i)(2-3i)} = \sqrt{25+9+4+9} \\ &= \sqrt{47} \end{aligned}$$

ଅବଶିଷ୍ଟ କର ।

$$9. u = (3-2i, 5i, 1+3i) = v, v = (5+2i, 2-3i, 7+4i)$$

$$u \cdot v = (3-2i)(5+2i) + 5i(2-3i) + (1+3i)(7+4i)$$

$$= (3-2i)(5-2i) + 5i(2+3i) + (1+3i)(7-4i)$$

$$= 15 - 6i - 10i + 4i^2 + 10i + 15i^2 + 7 - 4i + 21i - 12i^2$$

$$= 22 + 11i + 7i^2$$

$$= 22 + 11i - 7$$

$$= 15 + 11i$$

সাক্ষাৎ করে দেখাচ্ছে।

$$10. (i) u = (5, -2, -3, 7, -1)$$

$$\therefore \|u\| = \sqrt{25+4+9+49+1} = 2\sqrt{22}$$

$$\hat{u} = \frac{u}{\|u\|} = \left(\frac{5}{2\sqrt{22}}, -\frac{2}{2\sqrt{22}}, -\frac{3}{2\sqrt{22}}, \frac{7}{2\sqrt{22}}, -\frac{1}{2\sqrt{22}} \right)$$

$$(ii) v = (2, -1, 0, 7, 2)$$

$$\|v\| = \sqrt{4+1+0+49+4} = \sqrt{58}$$

$$\therefore \hat{v} = \frac{v}{\|v\|} = \left(\frac{2}{\sqrt{58}}, -\frac{1}{\sqrt{58}}, \frac{0}{\sqrt{58}}, \frac{7}{\sqrt{58}}, \frac{2}{\sqrt{58}} \right)$$